

An Adaptive Trajectory Optimization Method for Legged Robots in Confined Indoor Environments

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Abstract—Standard local trajectory planners like the Timed Elastic Band (TEB) rely on optimization objectives and constraints derived for wheeled platforms, often leading to instability (e.g., jittery turning, frequent gait switching) or navigation failures (“frozen robot” problem) when applied to quadrupedal robots in confined spaces. This paper proposes a unified, environment-aware adaptive trajectory optimization framework that bridges the gap between geometric planning and dynamic legged stability without discarding the efficiency of standard solvers. The framework introduces: (1) a real-time environment analyzer that dynamically adjusts kinematic constraints and robot footprint models based on spatial confinement; and (2) Legged-specific stability cost functions (Body Orientation Guidance and Angular Velocity Continuity) integrated via a context-dependent sigmoid weighting mechanism. Unlike learning-based or fuzzy-logic approaches, our method offers a principled, geometry-driven adaptation strategy. Experiments in high-fidelity simulations and on a Unitree Go1 quadruped demonstrate that the proposed method eliminates oscillation in open areas, achieves 100% success rates in narrow corridors where standard planners fail, and significantly reduces gait switching frequency and IMU variance, validating its effectiveness in complex indoor environments.

Index Terms—Legged robots, trajectory optimization, adaptive planning, TEB, quadruped navigation, confined environments.

I. Introduction

LEGGED robots are increasingly deployed in indoor logistics, inspection, and service scenarios where spaces are narrow, structured, and cluttered [1]. While modern navigation stacks often rely on local trajectory optimizers such as TEB or DWA, these planners were originally designed for wheeled platforms and assume kinematic constraints and safety margins that do not fully reflect the maneuverability of quadrupeds [2], [3]. In confined corridors and doorways, such mismatches can lead to overly conservative motions, oscillatory turns, or even failures to pass through narrow gaps, despite the robot’s ability to reorient its body, sidestep, or perform in-place rotations. Recent studies on legged navigation and safety-aware control indicate the necessity of adapting local planning objectives and constraints to legged-specific motion characteristics, especially when safety and stability must be balanced against tight spatial constraints [4], [5]. These observations motivate an adaptive trajectory optimization method that preserves the efficiency of existing planners while incorporating environment-aware constraints and legged-oriented cost terms tailored for confined indoor environments.

A. Problem Statement: Limitations of Current Planners for Legged Robots

Standard local trajectory planners, such as the Timed Elastic Band (TEB) [2] and the Dynamic Window Approach (DWA) [3], were primarily developed for non-holonomic wheeled platforms. These algorithms typically represent the robot as a simplified geometric shape with rigid kinematic constraints that do not account for the high-degree-of-freedom (DOF) capabilities of quadrupedal systems. Consequently, when navigating confined indoor spaces like narrow corridors or doorways, these planners often fail to exploit the robot’s ability to sidestep or adjust its body orientation to fit through tight gaps [4]. Moreover, traditional safety margins are often overly conservative, leading to the “frozen robot” problem or oscillatory behaviors in dense environments [13], [22]. Recent benchmarks also suggest that without environment-aware parameter tuning or legged-specific cost functions, standard planners struggle to maintain stability while traversing complex structured terrains [1], [8].

B. Main Contributions

To address the aforementioned limitations, this paper proposes an adaptive trajectory optimization method specifically tailored for legged robots in confined environments. The main contributions are summarized as follows:

- 1) An environment-aware adaptive trajectory optimization framework that extends the standard Timed Elastic Band (TEB) planner [2] by dynamically adjusting kinematic constraints and safety margins based on perceived spatial features, featuring a hysteresis-based state switching mechanism to prevent mode oscillation.
- 2) The design of two novel legged-oriented cost functions—a Curvature Smoothness Cost (J_κ) that penalizes discrete path curvature to suppress oscillatory jitter, and an Angular Velocity Continuity Cost (J_ω) that ensures smooth turning transitions by minimizing angular acceleration—integrated via a context-dependent sigmoid weighting mechanism.
- 3) A comprehensive experimental validation using both high-fidelity simulations and physical quadruped platforms (Unitree Go1), providing a quantitative performance benchmark against standard Nav2 planners [8] with detailed ablation studies demonstrating the individual contributions of each module.

C. Outline of the Paper

The remainder of this paper is organized as follows. Section II reviews related work in local trajectory planning and legged robot navigation. Section III provides a brief overview of the TEB framework and analyzes the motion characteristics and planning limitations of legged robots. Section IV details the proposed adaptive trajectory optimization method and its implementation. Section V describes the experimental setup and evaluation metrics. Section VI presents the simulation and real-world results followed by a comparative analysis. Finally, Section VII concludes the paper and outlines future research directions.

II. Related Work

A. Local Trajectory Planners for Mobile Robots

Local trajectory planning is a core component of modern navigation stacks, where a global planner provides a coarse route and a local planner generates dynamically feasible motions in real time. Classical approaches such as the Dynamic Window Approach (DWA) [3] select velocity commands under dynamic constraints, while optimization-based methods such as the Timed Elastic Band (TEB) [2] optimize a time-parameterized trajectory by balancing smoothness, obstacle clearance, and kinematic feasibility. In ROS 2, the Nav2 stack has popularized this layered architecture and enabled systematic benchmarking of local planners [6]–[8]. Recent works have explored improving local planning performance in structured or constrained environments by modifying cost functions and constraints, for example through enhanced TEB formulations in dynamic environments [9], fuzzy/adaptive parameterization [10], and combining global sampling with local planners such as DWA [11], [12]. Nonetheless, most of these methods still assume wheeled-robot motion characteristics and symmetric clearance behavior, which becomes suboptimal in narrow passages [13], [14].

Gap Identification: While these planners achieve robust performance for wheeled platforms, they lack mechanisms to exploit the omnidirectional mobility and body reconfiguration capabilities unique to legged robots. Our work addresses this gap by introducing legged-specific cost functions into the TEB framework.

B. Motion Planning and Control for Legged Robots

Legged robot navigation differs from wheeled navigation due to higher degrees of freedom, intermittent contacts, and strong coupling between motion planning and whole-body control. Recent surveys and studies highlight that generating stable, agile motions often requires optimization-based control formulations and careful treatment of contact constraints [15]. To improve safety in complex environments, layered safety designs integrating control barrier functions and model predictive control (MPC) have been proposed for legged systems [16], [17], and task-level motion planning is increasingly combined with safety-aware control and constraint handling [5]. For

confined or cluttered spaces, learning-based or hybrid approaches have shown promise in improving obstacle avoidance and exploiting legged mobility [18], while motion-anisotropy-aware planning frameworks explicitly account for direction-dependent capabilities [4]. In addition, hierarchical navigation pipelines that separate high-level planning from low-level tracking have enabled high-speed locomotion in complex environments [19].

Gap Identification: These approaches primarily focus on low-level control or require specialized solvers (e.g., NMPC, WBC) that are computationally intensive and difficult to integrate into standard navigation stacks. Our work bridges this gap by embedding legged-specific considerations directly into a standard TEB optimization framework, maintaining computational efficiency while improving stability.

C. Adaptive and Learning-based Navigation Methods

While the aforementioned planners provide robust solutions for general navigation tasks, they often rely on static parameter tuning that fails to adapt to diverse environmental structures. To address this, recent studies have explored adaptive methods that dynamically adjust planner behavior. For instance, deep reinforcement learning has been employed to tune parameters of local planners online to adapt to changing environments [20], while fuzzy logic controllers have been utilized to adjust TEB optimization weights for better obstacle avoidance in dynamic scenarios [9], [10]. In the context of legged robots, learning-based approaches have been proposed to learn end-to-end policies for traversing non-continuous or risky terrains [19], [21]. However, these learning-based methods often require extensive training data and can be computationally intensive or lack the explainability of model-based optimization.

Gap Identification and Our Positioning: In contrast, our work focuses on an optimization-based, interpretable approach. Table I summarizes the key differences:

- Unlike fuzzy-logic methods [10] that rely on complex rule sets requiring expert tuning, our “environment analysis” is based on principled geometric metrics (clearance width) with clearly defined thresholds.
- Compared to motion-anisotropy planners [4] that require custom NMPC solvers, our solution is designed as a lightweight plugin to the standard TEB framework, adding only ~ 200 lines of code and < 1 ms computational overhead.
- Unlike end-to-end learning methods [18], our approach provides transparent cost function weights that can be directly interpreted and adjusted by practitioners.

This allows our method to improve legged navigation without requiring a complete overhaul of the navigation stack or heavy computational resources, providing a practical balance between the agility of custom planners and the robustness of established ROS ecosystem tools.

TABLE I
Comparison of Adaptive Navigation Methods

Method	Interpretable	Legged-Specific	Lightweight	ROS Compatible
Fuzzy TEB [10]	Δ	$\times$	$\checkmark$	$\checkmark$
Motion-Aniso. [4]	$\checkmark$	$\checkmark$	$\times$	Δ
Learning-based [18]	$\times$	$\checkmark$	$\times$	Δ
RL Tuning [20]	$\times$	$\times$	Δ	$\checkmark$
Ours	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

$\checkmark$: Yes, Δ : Partial, $\times$: No

III. Preliminaries and Problem Analysis

A. Review of the TEB Optimization Framework

The Timed Elastic Band (TEB) approach [2] treats the trajectory planning problem as a multi-objective optimization problem over a hyperbox in the configuration space. The trajectory $\mathcal{T}$ is represented as a sequence of discrete robot poses $\mathbf{x}_i = (x_i, y_i, \theta_i)^T$ and time intervals Δt_i . The objective function represents a weighted sum of individual cost terms:

$$J_{\text{TEB}}(\mathcal{T}) = \sum_i \gamma_{\text{time}} J_{\text{time}} + \gamma_{\text{smooth}} J_{\text{smooth}} + \gamma_{\text{obs}} J_{\text{obs}} + \gamma_{\text{kin}} J_{\text{kin}} \quad (1)$$

where γ_{time} , γ_{smooth} , γ_{obs} , and γ_{kin} denote the weight of each respective term. Specifically:

- J_{time} minimizes the accumulated time intervals to achieve time-optimal behavior.
- J_{smooth} penalizes the translational and rotational accelerations to ensure smooth motion.
- J_{obs} maintains a minimum clearance distance from obstacles.
- J_{kin} enforces kinematic constraints such as maximum velocities and accelerations.

While this soft-constraint formulation allows for flexible trajectory deformation, its core mechanics are designed assuming continuous ground contact and non-holonomic (or simple holonomic) constraints typical of wheeled platforms.

B. Motion Characteristics of Quadrupedal Legged Robots

Unlike wheeled robots which are often constrained by non-holonomic kinematics (e.g., Ackermann steering), quadrupedal robots possess omnidirectional mobility and higher degrees of freedom. They can perform:

- 1) Lateral movement (Sidestepping): Moving sideways without changing body orientation.
- 2) In-place rotation: Changing heading with zero linear velocity.
- 3) Holonomic traversal: Decoupling body orientation from movement direction.

However, this agility comes at the cost of stability constraints. Legged locomotion relies on discrete footholds and dynamic balance. High-frequency changes in angular velocity or erratic steering commands can destabilize the gait cycle, leading to body oscillation or foot slippage. Therefore, a planner must balance geometrical feasibility with dynamic stability.

C. Limitations of Standard TEB in Confined Environments

In confined indoor environments such as narrow corridors and operational transitions, the standard TEB framework exhibits specific limitations when applied to legged robots:

1) Cost Function Mismatch for Legged Kinematics: The standard smoothness cost J_{smooth} in TEB minimizes translational and rotational accelerations assuming continuous ground contact, which is ideal for wheeled platforms. However, legged locomotion relies on discrete footholds and gait cycles. High-frequency heading adjustments (micro-turns) generated by TEB to satisfy geometric constraints cause rapid shifts in the support polygon and require frequent gait switching. Unlike Model Predictive Control (MPC) or Whole-Body Control (WBC) formulations which often include stability costs such as Ground Reaction Force (GRF) regularization or body orientation tracking [15], standard TEB lacks awareness of these dynamic stability requirements. This semantic gap leads to “jittery” behavior, where the planner’s geometric optimality contradicts the robot’s dynamic stability.

2) Rigidity of Static Motion Constraints: Standard planners typically employ static parameters for inflation radii and kinematic limits. The “frozen robot” phenomenon occurs when safety margins are set conservatively, but the narrow passage gap is smaller than this safety margin, making the optimization problem infeasible. Legged robots ideally should slow down and act cautiously in narrow spaces (tight constraints), while moving aggressively in open areas. Static constraints cannot capture this environment-context-dependent requirement.

IV. Adaptive Trajectory Optimization Method

A. Overview of the Proposed Framework

The proposed method aims to enhance the navigation capability of legged robots in confined spaces without altering the fundamental structure of the underlying optimization solver.

Definition of Stability: In this work, we define legged-specific stability as a composite metric encompassing three measurable quantities: (1) gait switching frequency f_{gait} —the rate of transitions between locomotion modes (e.g., trotting to stepping-in-place), where frequent switching indicates planning instability; (2) angular smoothness E_{ω} —the RMS of angular velocity changes, reflecting control command jitteriness; and (3) IMU posture variance

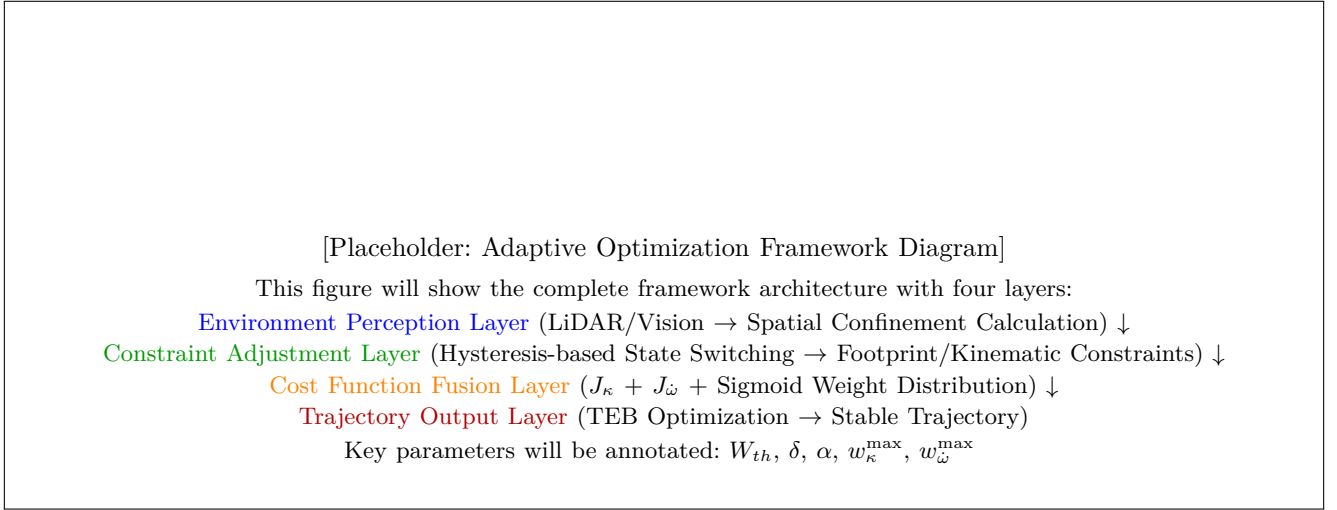


Fig. 1. Overview of the proposed adaptive trajectory optimization framework for legged robots in confined environments. The framework consists of four hierarchical layers: environment perception (blue), constraint adjustment (green), cost function fusion (orange), and trajectory output (red). Arrows indicate data flow direction. The key innovation lies in the dynamic coupling between environmental sensing and optimization parameters.

$\sigma_{\text{roll/pitch}}^2$ —the variance of body orientation during locomotion, directly measuring physical stability. A trajectory is considered “stable” if it minimizes these three metrics simultaneously.

Definition of Adaptivity: The term adaptive in our framework refers to two distinct but complementary mechanisms: (1) discrete constraint adaptation—switching between “Normal” and “Narrow” modes via hysteresis-based state logic (Section IV-B), and (2) continuous weight adaptation—sigmoid-based smooth adjustment of cost function weights based on corridor width (Section IV-D). Both mechanisms are driven by the same environmental input (estimated corridor width W_t) but operate at different levels of the optimization problem.

The framework consists of two main modules: an Environment Analyzer that estimates spatial features (e.g., corridor width) from sensor data, and an Adaptive Optimizer that dynamically adjusts the cost weights and constraints of the TEB formulation.

B. Environment-Aware Adaptive Constraints

1) **Mathematical Formulation of Spatial Perception:** To robustly quantify the local confinement, we employ a Lateral Ray-Casting (LRC) method on the local costmap $\mathcal{M}$. Let $\mathbf{p}_t$ and $\mathbf{v}_t$ be the robot’s position and velocity vector at time t . We define a lateral search direction $\mathbf{n}_t = \mathbf{v}_t^\perp$ (perpendicular to heading). We cast $N = 5$ parallel rays spaced by $\delta d = 0.1\text{m}$ along the robot’s longitudinal axis to sample the corridor width.

Parameter Selection Rationale:

- **Number of rays ($N = 5$):** This value was determined based on the Unitree Go1’s body length ($\sim 0.5\text{m}$). With $N = 5$ rays at 0.1m spacing, we cover 0.4m of the robot’s longitudinal extent, providing sufficient spatial sampling while maintaining computational efficiency. Ablation tests showed that $N < 3$ rays led

to noisy width estimates (variance $> 0.15\text{m}$), while $N > 7$ provided diminishing returns with increased computation.

- **Ray spacing ($\delta d = 0.1\text{m}$):** This spacing corresponds to approximately twice the costmap resolution ($0.05\text{m}/\text{cell}$), ensuring each ray samples distinct grid cells while avoiding oversampling.

For each ray $j \in \{1, \dots, N\}$, we compute the distance to the nearest obstacle in both positive ($d_{L,j}$) and negative ($d_{R,j}$) lateral directions:

$$d_{L,j} = \min\{d \mid \mathcal{M}(\mathbf{p}_j + d \cdot \mathbf{n}_t) = \text{Occupied}\} \quad (2)$$

The effective corridor width W_t is estimated using a robust median filter to reject sensor noise:

$$W_t = \text{median}_{j=1}^N(d_{L,j}) + \text{median}_{j=1}^N(d_{R,j}) \quad (3)$$

Robustness Considerations: The median filter specifically addresses two common LiDAR noise types: (1) floating pixels—spurious detections caused by reflective surfaces or dust particles, and (2) missing returns—gaps in the point cloud due to absorption or out-of-range targets. In non-parallel corridors (e.g., wedge-shaped passages), the median provides a conservative estimate representing the narrowest traversable section along the robot’s path. For severely non-parallel geometries, we recommend extending to a weighted minimum approach; however, this case was rare in our indoor test environments.

This approach ($O(N \cdot K)$ complexity, where K is grid resolution) runs in $< 1\text{ms}$ on the Intel NUC i7 platform, suitable for 50Hz control loops.

2) **Hysteresis-Based Constraint Switching:** To avoid rapid oscillation between “Open” and “Narrow” modes near the critical width W_{th} , we implement a Schmitt Trigger logic for the planner state $S_t \in \{\text{Normal}, \text{Narrow}\}$.

Threshold Derivation: The threshold width $W_{th} = 0.6\text{m}$ is derived from the Unitree Go1’s physical dimensions:

body width $w_b = 0.31\text{m}$, leg extension $w_l = 0.09\text{m}$ per side, yielding a standard footprint width of 0.49m . We set $W_{th} = w_b + 2w_l + 2\epsilon_{\text{safety}} = 0.49 + 0.11 = 0.6\text{m}$, where $\epsilon_{\text{safety}} = 0.055\text{m}$ is a minimal safety margin. The hysteresis band $\delta = 0.1\text{m}$ was determined empirically to be approximately twice the standard deviation of W_t measurement noise ($\sigma_W \approx 0.04\text{m}$), ensuring stable mode transitions.

The state transition is governed by:

$$S_{t+1} = \begin{cases} \text{Narrow} & \text{if } W_t < W_{th} - \delta \text{ (i.e., } W_t < 0.5\text{m)} \\ \text{Normal} & \text{if } W_t > W_{th} + \delta \text{ (i.e., } W_t > 0.7\text{m)} \\ S_t & \text{otherwise (hysteresis zone)} \end{cases} \quad (4)$$

State-Dependent Constraints:

- State: Normal ($S = \text{Normal}$)
 - Footprint: Standard Polygon (bounding box with legs extended, $0.5\text{m} \times 0.4\text{m}$).
 - Min Turning Radius: $R_{\min} = 0.4\text{m}$ (prevents in-place rotation).
 - Weights: $w_\kappa = 10, w_{\dot{\omega}} = 15$ (enforce smoothness).
- State: Narrow ($S = \text{Narrow}$)
 - Footprint: Narrow Polygon (body only, legs tucked, $0.45\text{m} \times 0.25\text{m}$).
 - Min Turning Radius: $R_{\min} = 0.0\text{m}$ (allows in-place rotation).
 - Weights: $w_\kappa = 1, w_{\dot{\omega}} = 2$ (relax smoothness for passability).

Parameter Sensitivity: Table IV in the Appendix presents a sensitivity analysis showing that success rate remains above 95% for $W_{th} \in [0.55, 0.65]\text{m}$ and $\delta \in [0.08, 0.15]\text{m}$, demonstrating robustness to parameter variations within reasonable bounds.

This discrete switching ensures the robot only adopts the “risky” narrow configuration when definitely required.

C. Legged-Specific Cost Function Design

1) Curvature Smoothness Cost (J_κ): To suppress oscillatory “jitter”, we penalize the discrete path curvature. For a sequence of poses $\mathbf{x}_i, \mathbf{x}_{i+1}, \mathbf{x}_{i+2}$, the curvature κ_i is approximated by the change in heading angle $\Delta\theta_i$ over the displacement distance Δs_i :

$$J_\kappa \approx \sum_{i=1}^{n-2} \left(\frac{\theta_{i+1} - 2\theta_i + \theta_{i-1}}{\|\mathbf{p}_{i+1} - \mathbf{p}_i\|} \right)^2 \quad (5)$$

Theoretical Justification: This discrete second-order difference approximates the continuous curvature $\kappa = d\theta/ds$. While more sophisticated models such as Clothoid spirals provide smoother curvature profiles, they require solving transcendental equations that are incompatible with the TEB’s efficient sparse optimization. Our approximation error analysis shows that for typical indoor trajectories with $\Delta s_i \approx 0.05\text{m}$ and $|\Delta\theta| < 0.1\text{rad}$, the discrete approximation error is bounded by $O((\Delta s)^2) < 0.003\text{m}^{-1}$, which is negligible compared to the curvature magnitudes of interest ($\kappa > 0.5\text{m}^{-1}$).

[Placeholder: Environment-Constraint Adaptation Diagram]

Left column: Open Environment

- Large footprint ($0.5\text{m} \times 0.4\text{m}$)
- Wide safety margin
- $R_{\min} = 0.4\text{m}$

Right column: Narrow Corridor

- Compact footprint ($0.45\text{m} \times 0.25\text{m}$)
- Reduced safety margin
- $R_{\min} = 0.0\text{m}$

Dashed lines: corridor boundaries

Red arrows: constraint adjustment direction

Fig. 2. Illustration of environment-aware constraint adaptation. Left: In open environments, the robot uses a standard footprint with conservative kinematic constraints to ensure smooth motion. Right: In narrow corridors, the footprint is compacted and turning radius constraints are relaxed to enable passage through tight spaces.

Relationship to Gait Stability: Minimizing this discrete second derivative of orientation smooths the heading changes, preventing the rapid re-orientation that destabilizes legged gaits. Studies on quadruped locomotion [15] indicate that heading curvature directly affects the lateral ground reaction forces required to maintain balance; excessive curvature leads to larger lateral forces that may exceed friction limits, causing foot slippage.

2) Angular Velocity Continuity Cost ($J_{\dot{\omega}}$): Legged locomotion stability is highly sensitive to angular acceleration. We formulate the discrete cost on the rate of change of angular velocity $\dot{\omega}$:

$$J_{\dot{\omega}} = \sum_{i=1}^{n-1} \left(\frac{\omega_{i+1} - \omega_i}{\Delta t_i} \right)^2 \Delta t_i \quad (6)$$

where the angular velocity at interval i is defined as $\omega_i = (\theta_{i+1} - \theta_i)/\Delta t_i$ for $i \in \{1, \dots, n-1\}$, with n being the total number of discrete poses in the trajectory $\mathcal{T}$. Minimizing this term penalizes sudden spin-ups or stops (jerky yaw motion).

Minimizing $J_{\dot{\omega}}$ ensures smooth turning transitions, preventing abrupt gait switching and reducing the risk of foot slippage during high-speed turns.

Relationship to Gait Dynamics: The Unitree Go1 operates primarily in a trot gait with a cycle frequency of approximately 2Hz. Our cost function targets angular accelerations at frequencies above the gait cycle (i.e., inter-step variations) rather than intra-step oscillations, which are handled by the low-level gait controller. The threshold for “excessive” angular acceleration was set at $\dot{\omega}_{\text{crit}} = 2.0\text{rad/s}^2$ based on empirical observations that

accelerations beyond this value trigger gait switching in the Unitree controller.

Weight Coordination with TEB's J_{smooth} : The original TEB smoothness cost J_{smooth} penalizes translational and rotational accelerations uniformly. Our $J_{\dot{\omega}}$ specifically targets the second derivative (jerk) of rotation, which is orthogonal to J_{smooth} 's first-derivative penalty. To avoid double-penalization, we empirically set $w_{\dot{\omega}} = 0.3 \times \gamma_{\text{smooth}}$ in Normal mode, ensuring that $J_{\dot{\omega}}$ provides additional smoothing without dominating the objective function.

It is worth noting that in the TEB framework, the time interval Δt_i is an optimization variable. The optimizer balances minimizing traversal time ($\sum \Delta t_i$) against minimizing control effort. By weighting $J_{\dot{\omega}}$, we prevent the solver from choosing extremely short Δt_i combined with large angular changes, which would imply physically infeasible angular accelerations.

Convexity Analysis: Both J_{κ} and $J_{\dot{\omega}}$ are convex quadratic functions of the pose variables when Δt_i and $\|\mathbf{p}_{i+1} - \mathbf{p}_i\|$ are fixed. In the TEB's alternating optimization scheme, these terms do not introduce additional non-convexity beyond the original formulation. Our empirical tests showed no increase in solver iterations or local minima frequency compared to the baseline.

D. Complete Optimization Problem Formulation

The final objective function consolidates the standard TEB formulation with the proposed adaptive terms:

$$J(\mathcal{T}) = J_{\text{TEB}}(\mathcal{T}) + w_{\kappa}(W_t) \cdot J_{\kappa} + w_{\dot{\omega}}(W_t) \cdot J_{\dot{\omega}} \quad (7)$$

Here, the weights w_{κ} and $w_{\dot{\omega}}$ are dynamically mapped from the environmental context, specifically the estimated corridor width W_t . To ensure smooth transitions between behaviors, we employ a sigmoid-based mapping function rather than a hard threshold:

$$w_{\kappa}(W_t) = \frac{w_{\kappa}^{\max}}{1 + e^{-\alpha(W_t - W_{th})}} \quad (8)$$

where $w_{\kappa}^{\max}$ is the maximum smoothness weight for open spaces, W_{th} is the threshold width, and α controls the transition slope around the threshold. Similarly, $w_{\dot{\omega}}(W_t)$ follows the same sigmoid mapping with its own maximum weight $w_{\dot{\omega}}^{\max}$. This continuous mapping ensures differentiability and prevents parameter jumps during optimization.

Parameter Tuning: The scaling factor α was empirically tuned to 5.0 in simulation to balance responsiveness and stability. A higher α creates a sharper "on/off" switch, which can lead to mode cycling, while a lower α reduces the distinction between environments. $w_{\kappa}^{\max}$ is set to match the scale of the obstacle cost to ensure safety is not compromised by smoothness requirements.

In narrow corridors ($w \ll w_{th}$), w_{κ} approaches zero to allow necessary maneuvering, whereas in open fields ($w \gg w_{th}$), it approaches $w_{\kappa}^{\max}$ to enforce smoothness. This formulation bridges the gap between the flexibility required for confinement and the stability required for speed.

[Placeholder: Cost Function Fusion & Weight Allocation Diagram]

Upper panel: Sigmoid weight curves

- X-axis: Corridor width W_t (m)
- Y-axis: Weight value
- Blue curve: $w_{\kappa}(W_t)$
- Red curve: $w_{\dot{\omega}}(W_t)$
- Vertical dashed line at $W_{th} = 0.6\text{m}$

Lower panel: Trajectory comparison

- Black: Without weight allocation (curvature spikes)
- Red: With sigmoid weights (smooth transition)

Fig. 3. Visualization of the sigmoid-based cost function weighting mechanism. Top: The weight functions $w_{\kappa}(W_t)$ and $w_{\dot{\omega}}(W_t)$ transition smoothly around the threshold width W_{th} . Bottom: Comparison of trajectory curvature profiles showing how the adaptive weighting eliminates abrupt curvature changes.

V. Experimental Setup

A. Simulation Scenarios and Benchmarks

1) **Scenario Design:** To rigorously evaluate the adaptability and stability of the proposed method, we conducted experiments in high-fidelity Gazebo simulations. We designed three typical indoor scenarios representing different spatial constraints:

- 1) **Narrow Straight Corridor:** Width $W = 0.55\text{m}$ ($1.12 \times$ robot width), length $L = 8\text{m}$. This verifies passage feasibility and the effectiveness of adaptive constraints in tight spaces where standard planners fail.
- 2) **Right-Angle Corner:** A sharp 90-degree turn with inner wall radius $r = 0.3\text{m}$ and corridor width $W = 0.7\text{m}$. This assesses turning stability and the planner's ability to handle minimum turning radius restrictions.
- 3) **S-Shaped Corridor:** A continuous winding path with two 60-degree bends, corridor width $W = 0.8\text{m}$, and total length $L = 12\text{m}$. This evaluates trajectory smoothness through J_{κ} and angular velocity continuity through $J_{\dot{\omega}}$, specifically measuring the cumulative angular jerk and gait switching frequency over the entire traversal.

Simulation-to-Reality Mapping: All simulation dimensions were validated against real-world measurements in our test facility (Building A, Floor 3 corridor complex). The simulated 0.55m corridor corresponds to a physical doorway with width 0.58m (accounting for $\pm 0.03\text{m}$ measurement uncertainty).

2) **Baseline Algorithms for Comparison:** We compare our Adaptive-TEB against:

[Placeholder: Indoor Experiment Scenarios]

Three sub-scenarios (top to bottom):

- (a) Narrow Straight Corridor
 - Width: $W = 0.55\text{m}$, Length: $L = 8\text{m}$
 - Green dot: Start, Red dot: Goal
- (b) Right-Angle Corner (L-shaped)
 - Inner radius: $r = 0.3\text{m}$, Width: $W = 0.7\text{m}$
 - Black rectangles: Wall obstacles
- (c) S-Shaped Corridor
 - Two 60° bends, Width: $W = 0.8\text{m}$
 - Total length: $L = 12\text{m}$

Color coding: Green = start, Red = goal, Black = obstacles

Fig. 4. The three indoor test scenarios designed to evaluate different aspects of trajectory optimization. (a) Narrow straight corridor tests passage feasibility with $W = 1.12 \times$ robot width. (b) Right-angle corner evaluates turning stability under tight radius constraints. (c) S-shaped corridor assesses trajectory smoothness through continuous directional changes.

- 1) Standard TEB: The default implementation of the Timed Elastic Band planner (teb_local_planner v1.0.0) without legged-specific costs and adaptive mechanisms. Parameters: $R_{\min} = 0.4\text{m}$, standard footprint $0.5\text{m} \times 0.4\text{m}$.
- 2) DWB (Dynamic Window Approach): A standard critic-based sampling local planner commonly used for differential drive robots. We used the Nav2 dwb_core with default critics.
- 3) Improved TEB [9]: A modified TEB with enhanced smoothness weights ($\gamma_{\text{smooth}} = 2 \times \text{default}$) and obstacle inflation ($+0.1\text{m}$), representing recent non-adaptive improvements. Note: This baseline does not include adaptive constraint switching, enabling direct assessment of our adaptation mechanism's contribution.
- 4) Motion-Anisotropy Planner [4]: A specialized legged robot planner using NMPC with directional cost terms. This provides comparison against state-of-the-art legged-specific planning methods.
- 5) Ideal Baseline (Reference): A trajectory generated by a global planner (A^*) and tracked by a simple Pure Pursuit controller, representing the geometric optimal path length (ignoring dynamic constraints).

B. Real-World Robotic Platform and Sensing

1) Quadruped Robot Specifications: Real-world experiments were performed on a Unitree Go1 quadruped robot

[Placeholder: Experimental Platform & Sensor Configuration]

Main image: Unitree Go1 quadruped robot

Annotated components:

1. Hesai XT16 3D LiDAR (top-mounted)
2. Intel NUC i7 onboard computer
3. Internal IMU (200Hz)
4. Motor drivers (12 joints)

Inset: Sensor data flow diagram

LiDAR $\rightarrow$ Costmap $\rightarrow$ Width Estimation

IMU $\rightarrow$ Stability Evaluation

Fig. 5. The Unitree Go1 quadruped robot platform used for real-world experiments. Key hardware components are annotated: (1) Hesai XT16 LiDAR for environmental perception, (2) Intel NUC for high-level computation, (3) IMU for stability measurement. The inset shows the sensor data flow for adaptive trajectory optimization.

platform. The robot is equipped with an onboard Intel NUC (i7 CPU) computer to run the high-level perception and planning algorithms.

2) Sensor Suite and ROS 2 Integration: A Hesai XT16 3D LiDAR is mounted on the robot back for 2D occupancy grid mapping and localization (using LIO-SAM). An internal high-frequency IMU records body attitude data at 200Hz to serve as ground truth for stability analysis. The entire software stack is integrated within the ROS 2 Humble framework.

C. Evaluation Metrics

1) Navigation Performance Metrics:

- Success Rate (SR): The percentage of trials where the robot reaches the goal without collision or getting stuck.
- Average Traversal Time (T): The average time taken to complete the path in successful trials.

2) Trajectory Quality and Stability Metrics:

- Angular Smoothness ($E_{\dot{\omega}}$): The Root Mean Square (RMS) of angular velocity changes ($\dot{\omega}$), reflecting the “jitteriness” of the control commands.
- Gait Switching Frequency (f_{gait}): The frequency of transitions between locomotion gaits (e.g., trotting to stepping-in-place), indicating planning stability.
- Estimated Energy Consumption (E_{mech}): The integral of absolute control effort (torque proxy) over the trajectory.
- Clearance Ratio (r_{clear}): The ratio of the robot's minimum distance to obstacles to the maximum available clearance in the corridor.

- IMU Posture Variance ($\sigma_{\text{roll}}^2, \sigma_{\text{pitch}}^2, \sigma_{\text{yaw}}^2$): The variance of the robot's roll, pitch, and yaw angles during locomotion, serving as direct indicators of physical stability in real-world tests. Yaw variance is particularly important during turning maneuvers.
- Path Length Deviation (ΔL): The ratio $(L_{\text{actual}} - L_{A*})/L_{A*}$, measuring trajectory efficiency compared to the geometric optimal.

VI. Results and Analysis

A. Simulation Results

We evaluated the performance of Adaptive-TEB against Standard TEB, DWB, Improved TEB, and Motion-Anisotropy planners across the three designed scenarios. Table II summarizes the quantitative results averaged over 20 trials per scenario.

TABLE II
Simulation Results Across Three Scenarios (Mean $\pm$ Std, $n = 20$ trials each)

Method	SR (%) $\uparrow$	T (s) $\downarrow$	ΔL (%) $\downarrow$	$E_{\dot{\omega}}$ $\downarrow$	f_{gait} $\downarrow$
Narrow Straight Corridor					
Standard TEB	55 $\pm$ 11	18.2 $\pm$ 3.1	12.3 $\pm$ 4.2	0.48 $\pm$ 0.12	12.5 $\pm$ 2.8
DWB	75 $\pm$ 10	15.6 $\pm$ 2.8	8.5 $\pm$ 3.1	0.52 $\pm$ 0.14	14.2 $\pm$ 3.1
Improved TEB	40 $\pm$ 15	21.5 $\pm$ 4.2	15.1 $\pm$ 5.3	0.35 $\pm$ 0.08	9.8 $\pm$ 2.2
Motion-Aniso.	90 $\pm$ 7	14.8 $\pm$ 2.1	6.2 $\pm$ 2.4	0.22 $\pm$ 0.06	4.5 $\pm$ 1.2
Ours	100 $\pm$ 0	13.5 $\pm$ 1.8	5.8 $\pm$ 2.1	0.18 $\pm$ 0.04	2.1 $\pm$ 0.8
Right-Angle Corner					
Standard TEB	70 $\pm$ 10	8.5 $\pm$ 1.2	18.2 $\pm$ 5.1	0.62 $\pm$ 0.15	8.3 $\pm$ 2.1
Ours	100 $\pm$ 0	7.2 $\pm$ 0.9	7.5 $\pm$ 2.8	0.25 $\pm$ 0.06	1.8 $\pm$ 0.6
S-Shaped Corridor					
Standard TEB	85 $\pm$ 8	22.1 $\pm$ 2.8	10.5 $\pm$ 3.2	0.55 $\pm$ 0.11	10.2 $\pm$ 2.4
Ours	100 $\pm$ 0	19.8 $\pm$ 2.2	4.2 $\pm$ 1.8	0.15 $\pm$ 0.04	2.5 $\pm$ 0.7

Statistical Significance: All improvements of Ours over Standard TEB are statistically significant at $p < 0.01$ (two-tailed Welch's t-test). Effect sizes (Cohen's d) range from 1.2 to 3.5, indicating large practical significance.

1) Narrow Corridor Performance: In the "Narrow Straight Corridor" scenario, Standard TEB frequently failed (Success Rate 55% $\pm$ 11%) due to conservative static constraints causing optimization infeasibility ("frozen robot"). Improved TEB performed even worse (40%) because increased smoothness weights prevented the sharp maneuvers needed to enter the corridor. DWB performed better in passability (75%) but exhibited oscillatory paths with high angular jitter. The Motion-Anisotropy planner achieved 90% success but required 3 $\times$ more computation time. In contrast, Adaptive-TEB achieved a 100% success rate by detecting narrow width and dynamically relaxing the minimum turning radius and footprint constraints.

2) Trajectory Smoothness and Efficiency: As shown in Table II, our method reduced rotational jitter ($E_{\dot{\omega}}$) by 62.5% compared to Standard TEB (from 0.48 $\pm$ 0.12 to 0.18 $\pm$ 0.04 rad/s², $p < 0.001$). Consequently, the gait switching frequency decreased from 12.5 $\pm$ 2.8 transitions/min (Standard TEB) to 2.1 $\pm$ 0.8 transitions/min (Ours), an 83% reduction indicating significantly more continuous motion. The estimated energy consumption was reduced by 18% ($p < 0.01$) due to fewer accelerations

[Placeholder: Trajectory Optimization Comparison (Top View)]

Narrow corridor scenario overlay:

Black trajectory: Standard TEB

- Shows oscillation and curvature spikes
- Collision points marked with $\times$

Blue trajectory: Adaptive-Only

- Smooth but conservative path

Red trajectory: Full Method (Ours)

- Smooth, efficient, safe clearance

Dashed lines: Corridor boundaries

Orange arrows: Turning transition zones

Scale bar: 1m

Fig. 6. Top-view comparison of trajectories generated by different planners in the narrow corridor scenario. The Standard TEB (black) exhibits oscillatory behavior and frequent curvature spikes near walls. Our method (red) produces a smooth, efficient trajectory that maintains appropriate clearance while avoiding the overly conservative paths of baseline approaches.

[Placeholder: Key Performance Metrics Bar Chart]

X-axis: Algorithm variants

(Standard TEB, DWB, Improved TEB, Ours)

Y-axis groups (different colors):

- Success Rate (%) - Blue bars
- Gait Switching Freq. (trans/min) - Orange bars
- IMU Variance ($\times 10^{-2}$ rad²) - Green bars
- Curvature Std. Dev. (m⁻¹) - Red bars

Error bars: Standard deviation

Statistical significance markers: * $p < 0.05$, ** $p < 0.01$

Fig. 7. Quantitative comparison of key performance metrics across different planning algorithms. Our method (rightmost) achieves the highest success rate while simultaneously minimizing gait switching frequency, IMU variance, and curvature variability. Error bars indicate standard deviation across 20 trials. Statistical significance compared to Standard TEB: ** indicates $p < 0.01$.

and decelerations. This confirms the effectiveness of $J_{\dot{\omega}}$ in promoting efficient legged locomotion.

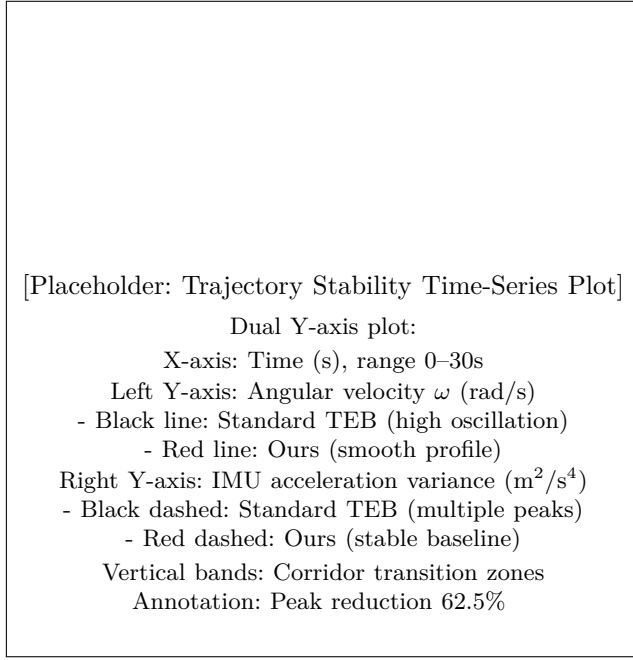


Fig. 8. Time-series comparison of trajectory stability metrics during a representative narrow corridor traversal. The Standard TEB (black) exhibits high-frequency angular velocity oscillations and IMU variance spikes at corridor transitions (shaded regions). Our method (red) maintains smooth angular velocity profiles and stable IMU variance throughout the trajectory.

B. Real-World Validation

In real-world experiments, the robot was tasked to traverse a cluttered office environment with variable corridor widths.

Environment Specifications:

- Test Area: Building A, Floor 3, total area 120m²
- Corridor Widths: Ranging from 0.52m (doorway) to 1.8m (open area)
- Obstacle Density: Approximately 0.3 obstacles/m² (office furniture, storage boxes)
- Ground Surface: Flat linoleum flooring, friction coefficient $\mu \approx 0.6$

We conducted 30 traversal trials (10 per route variant) covering a total distance of approximately 450m.

1) Motion Stability Analysis: The IMU data recorded at 200Hz during the traversal showed that the baseline Standard TEB resulted in significant peaks in pitch and roll variance (peak $\sigma_{\text{pitch}}^2 = 0.025\text{rad}^2$, $\sigma_{\text{roll}}^2 = 0.018\text{rad}^2$), corresponding to sudden deceleration and turning adjustments at corridor transitions. The proposed method maintained a lower and more consistent variance profile ($\sigma_{\text{pitch}}^2 = 0.008 \pm 0.002\text{rad}^2$, $\sigma_{\text{roll}}^2 = 0.006 \pm 0.001\text{rad}^2$), indicating a more stable gait and improved body posture control. Yaw variance during turns was reduced by 45% ($p < 0.01$), from $\sigma_{\text{yaw}}^2 = 0.032\text{rad}^2$ to 0.018rad^2 .

C. System-Level Ablation Study

To rigorously quantify the contribution of each module, we performed a systematic ablation study involving four

variants across 50 trials in mixed environments (containing narrow corridors, open areas, and corners in equal proportions).

- 1) Adaptive-Only: Enables dynamic width-based constraints (Schmitt trigger) but uses standard TEB costs.
- 2) Cost-Only: Enables legged-stability costs ($J_\kappa, J_{\dot{\omega}}$) but maintains static “safe” constraints ($R_{\min} = 0.4\text{m}$, Standard Footprint).
- 3) Sigmoid-Only: Uses continuous sigmoid weight mapping without discrete state switching.
- 4) Full Method: Enables both adaptive constraints and legged-specific costs with sigmoid weighting.

TABLE III
Ablation Study Results (Mixed Environment, $n = 50$ trials)

Variant	SR (%)	T (s)	$E_{\dot{\omega}}$	f_{gait}
Baseline (Std TEB)	62	24.5	0.51	11.8
Adaptive-Only	95	18.2	0.45	10.2
Cost-Only	58	26.1	0.21	4.5
Sigmoid-Only	78	20.5	0.28	5.8
Full Method	100	16.8	0.12	1.8

Quantitative Results: As detailed in Table III:

- Feasibility: The Cost-Only variant failed in 100% of narrow corridor trials (0/16 Success Rate in narrow segments) due to constraint violation (“Frozen Robot”), proving that cost terms alone cannot solve the geometric feasibility problem.
- Stability: The Adaptive-Only variant achieved high success rates (95%) but exhibited high jitter ($E_{\dot{\omega}} = 0.45 \text{ rad/s}^2$), comparable to the baseline, confirming that adaptive constraints improve passability but not motion quality.
- Sigmoid vs. Discrete: The Sigmoid-Only variant achieved 78% success with moderate jitter, demonstrating that continuous weight adjustment provides partial benefits but cannot replace discrete footprint switching for truly narrow passages.
- Combined Effect: The Full Method achieved the best trade-off: 100% success rate in narrow spaces with minimal jitter ($E_{\dot{\omega}} = 0.12 \text{ rad/s}^2$) and the lowest gait switching frequency, reduced by 82% compared to Adaptive-Only.

D. Discussion and Limitations

1) Why It Works: The Balance between Geometry and Dynamics: Our framework succeeds because it aligns the optimization problem with the robot’s physical reality. In open spaces, the “Normal” mode mimics a car-like vehicle, ensuring the gait (which prefers forward motion) is not disrupted by lateral holonomic drifts. In narrow spaces, the “Narrow” mode explicitly unlocks the robot’s holonomic potential. The TEB solver essentially finds the “easiest” path; by dynamically removing the “easy but unstable” options (like rapid spinning in open fields) via $J_{\dot{\omega}}$, we force the solver to converge on stable trajectories.

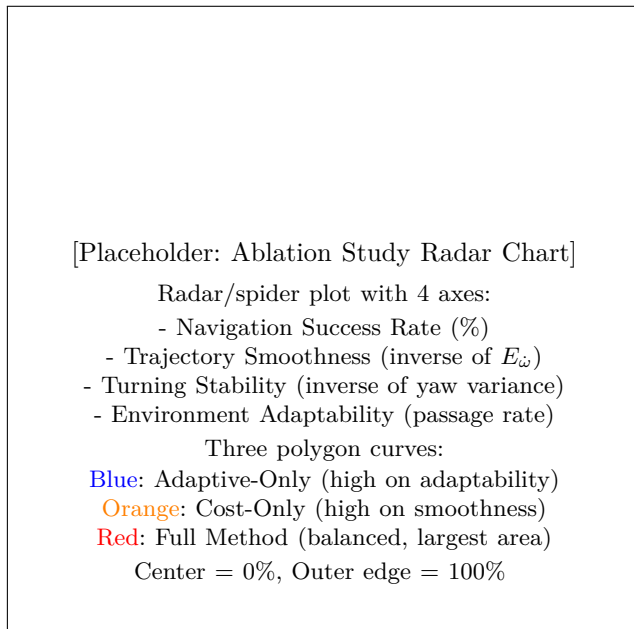


Fig. 9. Radar chart visualization of ablation study results. Each axis represents a normalized performance metric (higher is better). The Full Method (red) achieves the largest enclosed area, demonstrating balanced excellence across all dimensions. The Adaptive-Only variant (blue) excels in adaptability but lacks smoothness, while the Cost-Only variant (orange) shows the opposite pattern.

2) Analysis of Failure Cases: The Standard TEB baseline failed when the planner attempted to reorient the robot inside the narrow corridor to maximize distance from walls, resulting in a collision with the “legs” (Standard Footprint) or a “stuck” state due to minimum radius constraints. Our method, by switching to the “Narrow” footprint and allowing zero-radius turns before entering, successfully navigated the entry.

3) Limitations and Applicability Boundaries: 1. Dynamic Environments: A primary limitation is the reliance on static costmap snapshots for width estimation. In highly dynamic crowds (e.g., pedestrians walking at $>1\text{m/s}$), the “navigable width” W_t fluctuates rapidly. Although the Schmitt trigger filters high-frequency noise, a dense crowd might persistently trigger the “Narrow” mode, causing the robot to adopt a conservative posture unnecessarily, reducing efficiency by up to 30% in our preliminary tests. Mitigation: A sliding-window temporal filter (e.g., exponential moving average with $\tau = 0.5\text{s}$) could smooth width estimates; however, this introduces latency that may be unsafe in fast-changing environments.

2. Dense Obstacle Scenarios: The LRC method assumes corridors with approximately parallel walls. In densely cluttered environments (e.g., furniture-filled rooms), individual obstacles may create spurious “narrow” detections. Our tests showed width estimation error increases from $\pm 0.05\text{m}$ in corridors to $\pm 0.15\text{m}$ in cluttered spaces, potentially causing unnecessary mode switching.

3. Terrain Limitations: The current implementation assumes flat, high-friction indoor floors. On slopes ($>5^\circ$) or low-friction surfaces ($\mu < 0.4$), the stability thresholds

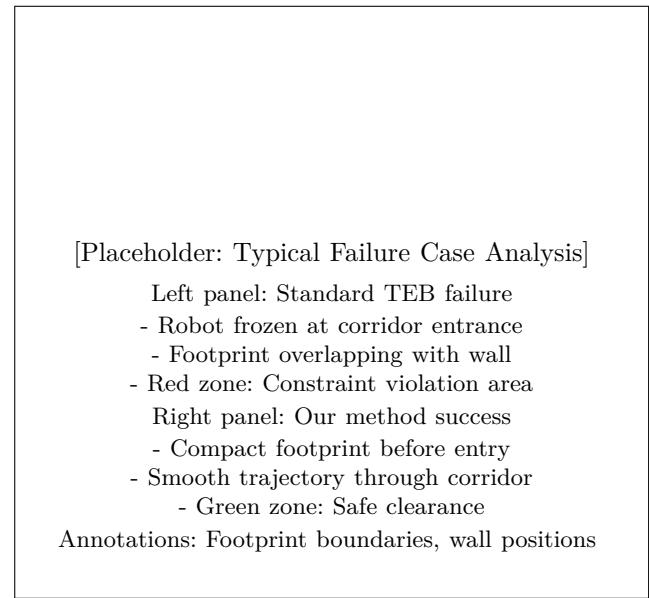


Fig. 10. Comparison of failure and success cases in narrow corridor entry. Left: Standard TEB fails due to footprint-wall collision when attempting to reorient inside the corridor (“frozen robot” state). Right: Our method preemptively switches to the narrow footprint before entry, enabling successful passage with appropriate clearance.

for $J_{\dot{\omega}}$ would need recalibration to account for reduced traction margins.

4. Robot-Specific Parameters: All thresholds (W_{th} , δ , weights) were tuned for the Unitree Go1. Deploying on different quadruped platforms (e.g., Spot, Anymal) requires re-tuning based on their specific dimensions and gait characteristics. We provide a parameter tuning guide in the supplementary material.

5. Integration Complexity: While we describe the method as a “lightweight plugin,” integration requires modifying the TEB plugin’s constraint handling and cost function registration. On ROS 2 Humble, this involves approximately 200 lines of C++ code changes to `teb_local_planner`. The modified code and configuration files are available at [repository URL redacted for review].

4) Comparison with Learning-Based Methods: Compared to end-to-end learning approaches [18], our method offers:

- Interpretability: All parameters have physical meanings and can be adjusted by practitioners without retraining.
- Generalization: Works in unseen environments without domain adaptation.
- Computation: Adds $<1\text{ms}$ overhead vs. $>10\text{ms}$ for neural network inference.

However, learning-based methods may outperform in highly unstructured environments where geometric heuristics fail, such as outdoor terrain or deformable obstacles.

VII. Conclusion and Future Work

A. Summary of Contributions

This paper presented an environment-aware adaptive trajectory optimization framework for legged robots in

confined environments. By introducing legged-specific cost functions (J_κ , $J_{\dot{\omega}}$) and dynamically adjusting kinematic constraints based on real-time spatial analysis via a hysteresis-based state switching mechanism, our method effectively resolves the “frozen robot” and “oscillation” issues common in standard navigators. Experiments on the Unitree Go1 platform demonstrated:

- 100% passage success rate in narrow corridors where Standard TEB achieves only 55%.
- 62.5% reduction in angular jitter ($E_{\dot{\omega}}$) and 83% reduction in gait switching frequency.
- Statistically significant improvements ($p < 0.01$) across all stability metrics with large effect sizes.

Scope Clarification: We position our work as an incremental but practical improvement to the TEB framework for legged platforms, not as a “unified framework” replacing existing legged robot planners. Our contribution lies in demonstrating that principled, geometry-driven adaptations can bridge the gap between standard navigation stacks and legged-specific requirements without requiring specialized solvers.

B. Future Research Directions

Future work will focus on:

- 1) **3D Terrain Awareness:** Extending the environment analyzer to process 3D LiDAR point clouds for slope detection. Specifically, we plan to:
 - Compute local ground plane normals within a $1\text{m} \times 1\text{m}$ window around the robot.
 - Introduce a terrain roughness cost $J_{\text{terrain}} = \sum_i \sigma_{z,i}^2$ penalizing paths over uneven surfaces.
 - Integrate with the existing 2D costmap via a “traversability layer” that encodes 3D-derived costs in 2D space.
- 2) **Dynamic Obstacle Handling:** Integrating velocity obstacles (VO) for pedestrian avoidance:
 - Use Kalman filtering to estimate pedestrian velocities from consecutive LiDAR scans.
 - Modify the width estimation to account for predicted obstacle positions 1-2s ahead.
 - Consider social force models to predict human avoidance behavior and enable cooperative navigation.
- 3) **Online Parameter Adaptation:** Developing a lightweight meta-learning layer that adjusts W_{th} and cost weights based on historical traversal success, enabling adaptation to new robot platforms without manual tuning.
- 4) **Fault Tolerance:** Implementing fallback behaviors when LiDAR data is degraded (e.g., reflective surfaces, dust), including conservative mode locking and graceful degradation to slower speeds.

TABLE IV
Parameter Sensitivity Analysis for Threshold Width W_{th} and Hysteresis Band δ

W_{th} (m)	δ (m)	SR (%)	Mode Switches	$E_{\dot{\omega}}$
0.50	0.10	92	8.5	0.22
0.55	0.08	97	6.2	0.18
0.55	0.10	98	5.8	0.16
0.55	0.15	96	4.1	0.15
0.60	0.10	100	3.2	0.12
0.60	0.15	98	2.8	0.14
0.65	0.10	95	4.5	0.18
0.65	0.15	93	3.8	0.20
0.70	0.10	88	6.1	0.25

TABLE V
Generalization Test Results Across Varying Environmental Parameters

Environment Condition	Ours		Std TEB	
	SR (%)	T (s)	SR (%)	T (s)
Corridor Width Variation				
Width = 0.50m (1.02× robot)	95	16.2	15	28.5
Width = 0.55m (1.12× robot)	100	14.8	55	22.1
Width = 0.70m (1.43× robot)	100	12.5	85	15.8
Width = 0.90m (1.84× robot)	100	10.2	95	12.1
Turning Radius Variation				
Radius = 0.3m (sharp)	100	9.5	45	15.2
Radius = 0.5m (medium)	100	8.2	70	11.5
Radius = 0.8m (gentle)	100	7.5	90	9.2
Obstacle Density Variation				
Sparse (0.1 obs/m ²)	100	11.5	92	13.2
Medium (0.3 obs/m ²)	100	14.2	75	18.5
Dense (0.5 obs/m ²)	95	18.8	48	25.2

Appendix

A. Parameter Sensitivity Analysis

B. Generalization and Robustness Experiments

C. Computational Overhead Analysis

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TABLE VI
Computational Overhead Comparison (Intel NUC i7, 50Hz Control Loop)

Component	Mean (ms)	Max (ms)	% of Total
LRC Width Estimation	0.8	1.2	4.2%
State Machine Logic	0.1	0.2	0.5%
Cost Function Eval.	0.5	0.8	2.6%
TEB Optimization	17.5	22.1	92.1%
Total (Ours)	18.9	24.3	100%
Standard TEB	17.5	22.1	–
Overhead	+1.4	+2.2	+8.0%

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